Jonathan Engelhard

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github.com/JoNoToPo

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, C, Kotlin, SQL, HTML, CSS
AI/ML: Machine learning, MediaPipe hand tracking, OpenAI API (DALL-E 3)
Frameworks: React, Vite, Node.js, Express, MongoDB, Tone.js, Web MIDI, Pyodide, JavaFX
Tools: Git, GitHub Actions & Pages, Claude Code, Gradle, WordPress, AppSheet, Firebase, Figma

INDUSTRY PROJECTS

HIIPS360 - WorkSafeBC *(Industry Sponsored Project, Client Liaison)* Sep 2026 - present

- Building a prototype PWA with a team of 6 that turns WorkSafeBC's 20+ page hazard inspection PDFs into searchable, regulation-linked guidance for officers on site
- Act as client liaison with WorkSafeBC; built the job sites home page and /jobs flow in React
- Technologies: React, Vite, Express, MongoDB, Google Gemini API

Autism Community Training (ACT) *(Practicum, Client Liaison)* Apr 2026

- Served as the team's main client liaison with ACT, a B.C. autism training non-profit, turning staff requests into changes on their WordPress staging site
- Rebuilt the navbar, home and workshop pages; added Courses and Family Navigation pages
- Removed outdated pages and courses; updated resources and New Diagnosis Hub content
- Added the Visibility Control for LearnDash and What The File plugins

PERSONAL PROJECTS

Synth Conductor | Live demo Oct 2026 - present

- Built a webcam-controlled browser instrument, inspired by Gesture Synth (with its creator's permission): one hand picks chords, the other shapes sound
- Ran MediaPipe hand tracking fully in-browser, so camera video never leaves the device
- Synthesized audio with Tone.js, added Web MIDI output, a looper and a guided tutorial, and deployed with GitHub Actions to GitHub Pages

ACADEMIC PROJECTS

Rewrite - BCIT Python Course | Play in browser Mar 2025 - Nov 2026

- Created a time-travel puzzle game about rewriting history, with procedural ASCII graphics
- Ported it to the browser with Pyodide, added saved history between runs, wrote 178 unit tests

Paradax - BCIT Java Course Sep 2025 - Dec 2025

- Built a JavaFX puzzle game solved by resizing and moving the game window, over 16 levels
- Applied SOLID design (Level interface for dependency inversion), added auto-save, and enforced code quality with Checkstyle, SpotBugs and PMD in Gradle

Gamify Life - BCIT Projects Course May 2025 - Jun 2025

- Led a team of 5 using Scrum to build a web app that lets parents gamify chores, with AI-generated logos from the DALL-E 3 API (MongoDB, Express, Node.js)

EXPERIENCE

Energy Advisor - Energuy Jan 2021 - Aug 2024

- Earned several 5-star reviews, with customers emailing management to praise my service

Volunteer Leader - Church of Zion May 2018 - present

- Help lead Tuesday worship, the Saturday young adults group and the Sunday dance team

EDUCATION

Computer Systems Technology Diploma Expected Dec 2026
British Columbia Institute of Technology, Burnaby, BC
AI & Machine Learning Option: Machine Learning, Intro to AI, AI/ML Topics

Acting Diploma Graduation: Sept 2020
New Image College, Vancouver, BC